

# **From Price to Ratings: A Multi-Source Analysis of Online Electronics Products**

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DSCI 510 Final Project**

# Introduction

Project Goal:

- Analyze how product price, review count, data source, and subcategory relate to customer ratings.

Research Questions:

- Are more expensive products rated higher?
- Are products with more reviews rated higher?
- Do different data sources show different product patterns?
- Can price, review count, source, and subcategory predict rating?

# Data Sources and Final Analysis Scope

| Source Name                    | Type          | Final Rows | Main Fields                                                                                                                   | Role in Project        |
|--------------------------------|---------------|------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------|
| All_Electronics.csv            | CSV           | 8,233      | product name, price, rating, review count, category                                                                           | main analysis          |
| Home_Entertainment_Systems.csv | CSV           | 7,292      | product name, price, rating, review count, category                                                                           | main analysis          |
| Amazon Best Sellers webpages   | Web scraping  | 298        | product name, price, rating, review count                                                                                     | Best-seller comparison |
| DummyJSON Products API         | API           | 193        | product name, price, rating, review count, category                                                                           | API-based comparison   |
| Total                          | Mixed sources | 16,016     | product name, price, rating, review count, category, subcategory, souce, source label, data role, log price, log review count | Final analysis sataset |

# Feature Description

| Feature          | Description                        | Measurement                      |
|------------------|------------------------------------|----------------------------------|
| price            | Product listed price               | Numeric value in USD-like format |
| rating           | Customer product rating            | 0-5 scale                        |
| review_count     | Number of customer reviews/ratings | Count variable                   |
| log_price        | Log-transformed price              | $\log(1 + \text{price})$         |
| log_review_count | Log-transformed review count       | $\log(1 + \text{review\_count})$ |
| source           | Data origin                        | CSV / API / web scraping         |
| subcategory      | Product group                      | Product category label           |

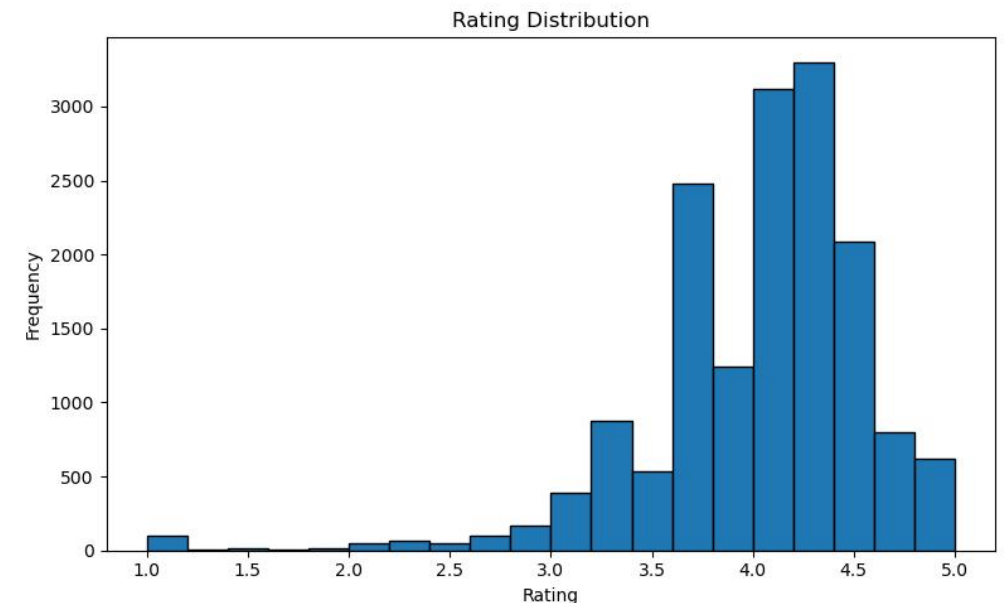
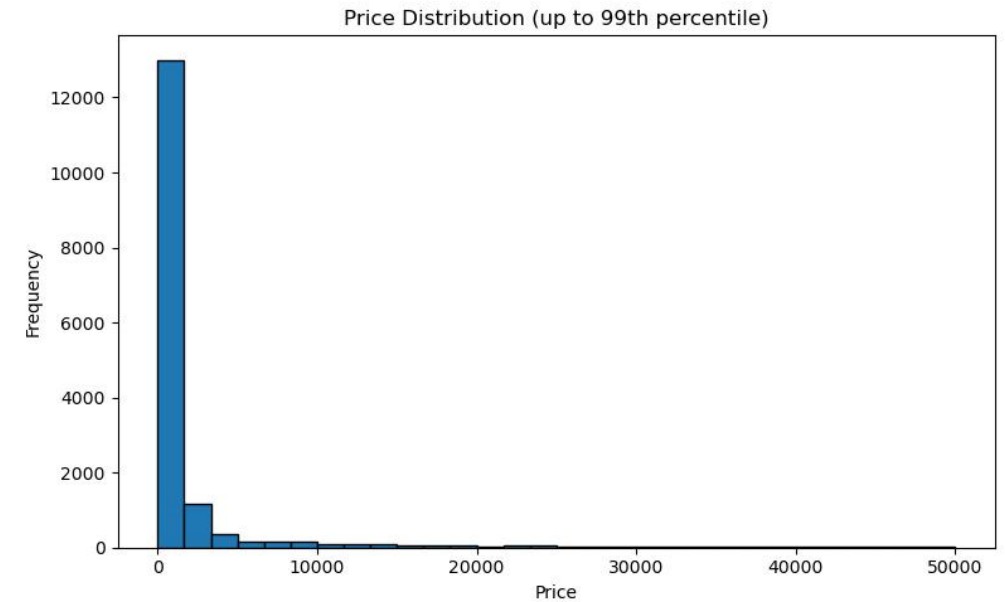
# Summery of the results

## Final Analysis Dataset:

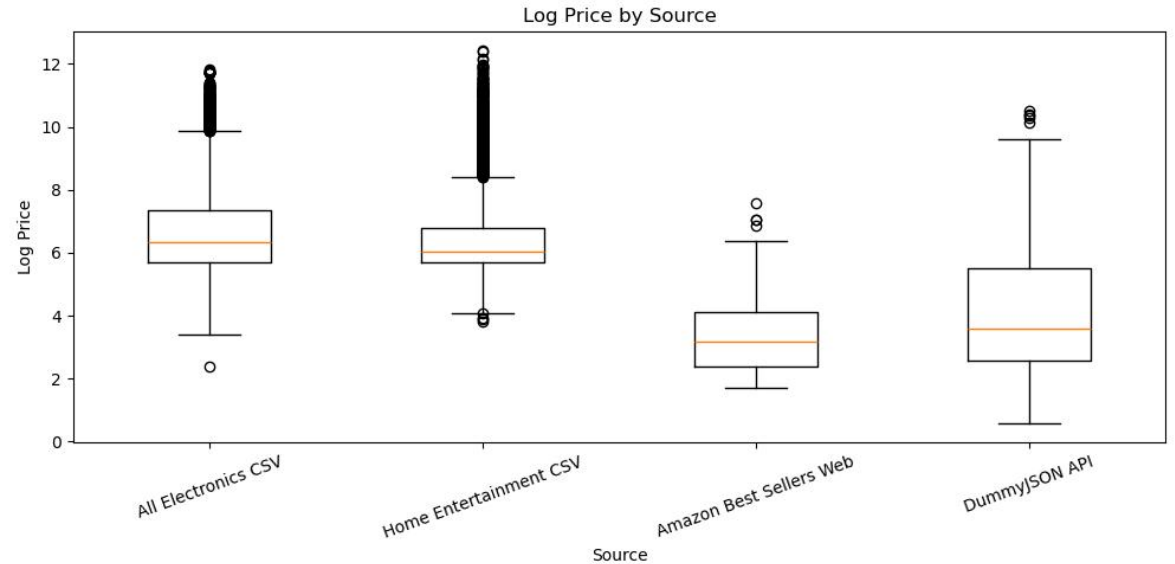
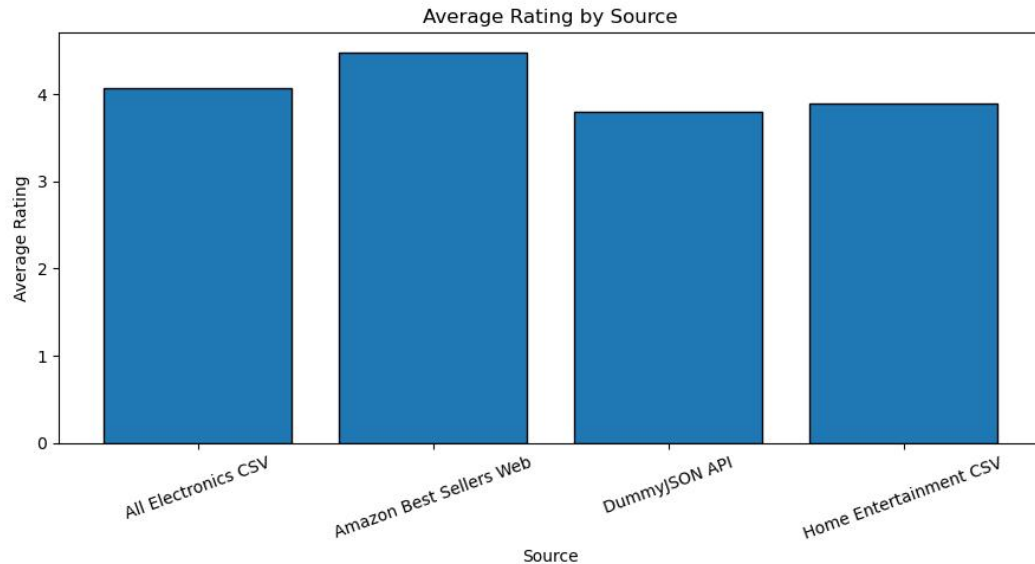
- 16,016 products from 4 sources
- 9 subcategories
- Mean price = 2,859.50; median price = 449.00
- Mean rating = 4.00; median rating = 4.10
- Mean review count = 6,022.28; median review count = 280

## Main Finding:

Price and review count are strongly right-skewed,  
while ratings are concentrated near 4.0–4.5.



# Source Comparison Results



## Key Source-Level Patterns:

- Amazon Best Sellers has the highest mean rating: 4.48
- DummyJSON has lower average rating: 3.80
- All Electronics has the highest mean price among main Amazon CSV sources: 3,066.67
- Amazon Best Sellers has much lower median price: 22.60
- Amazon Best Sellers has the highest mean review count: 33,497.29

# Correlation and Statistical Testing

## Correlation Results:

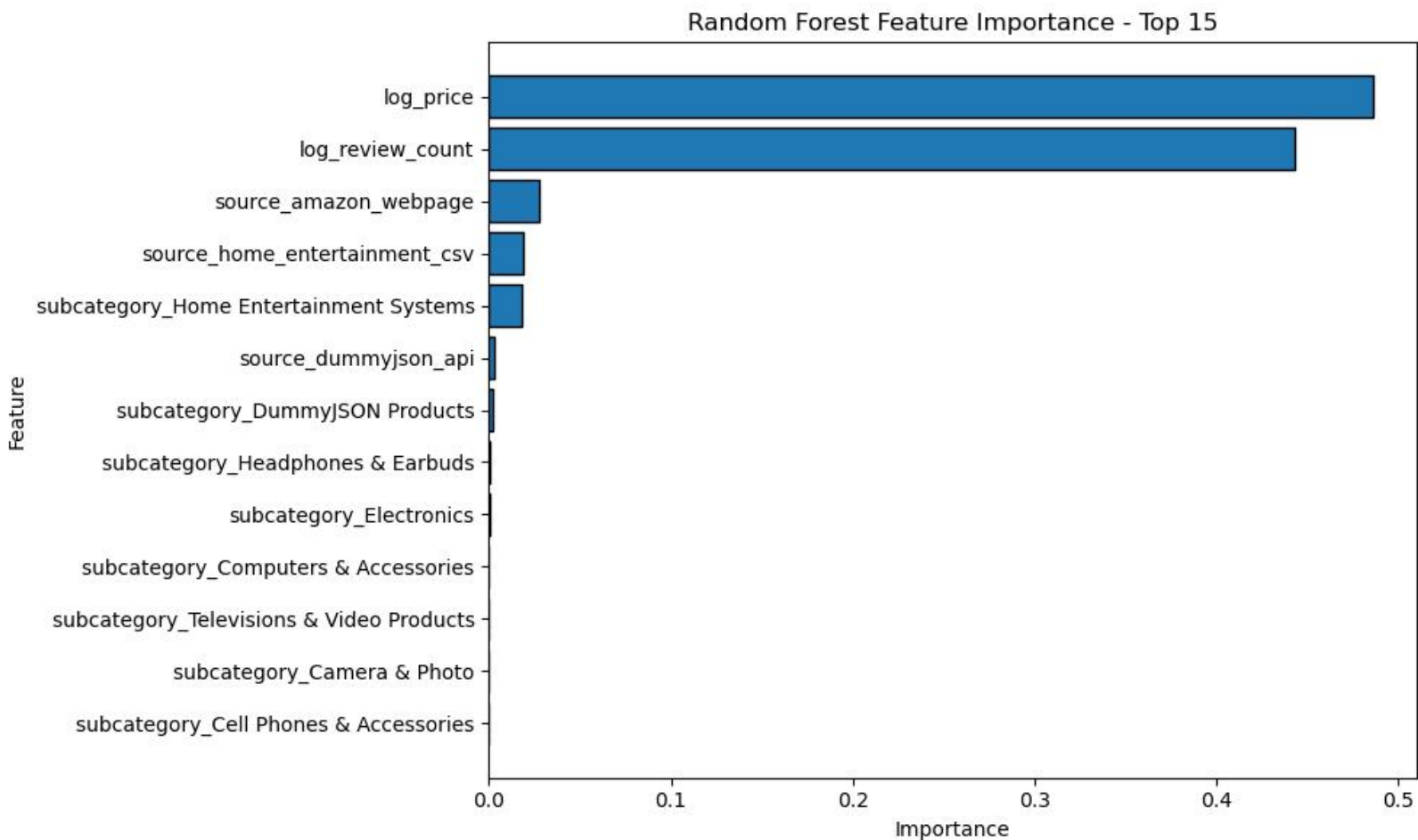
- Price and rating have a weak positive relationship.  
Pearson  $r = 0.0905$ ; Spearman  $r = 0.0950$
- Review count and rating also have a weak positive relationship.  
Pearson  $r = 0.0958$ ; Spearman  $r = 0.2271$
- Log review count has the strongest relationship with rating among these variables.
- Pearson  $r = 0.2467$

## Hypothesis Testing:

- Source-level differences are statistically significant in many comparisons.
- Best Sellers products have higher average rating than the main Amazon CSV products.

# Regression and Machine Learning Results

| Model         | Target | Rows Used | MAE   | RSME  | R <sup>2</sup> |
|---------------|--------|-----------|-------|-------|----------------|
| Random Forest | rating | 16,016    | 0.344 | 0.506 | 0.096          |



Interpretation:  
Ratings cannot be strongly predicted by price, review count, source, and subcategory alone. Other factors such as brand reputation, product quality, customer expectations, and shipping experience may matter.

# Challenges

- Different sources used different schemas, so I had to standardize names, prices, ratings, and review counts.
- Amazon web scraping was unstable because Amazon pages may block scraping or change HTML structure.
- The four data sources had different collection methods, so direct comparison required caution.
- Ratings were highly concentrated near 4.0–4.5, which made prediction difficult.
- Some important rating factors, such as brand reputation and product quality, were not included in the data.

**Thank you!**